AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

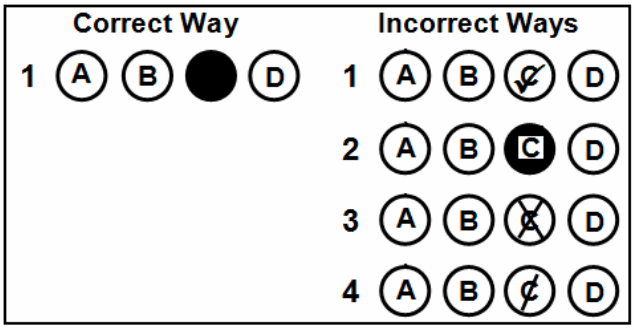
MAY 2013

General Mathematics Paper I

Time: 40 minutes Marks: 30

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. If 3% of $x = 3$, then 10% of x is

   A. 1
   B. 9
   C. 10
   D. 100

2. If $\frac{x-1}{2(y-1)} = \frac{1}{2}$, then $x : y$ is equal to

   A. 1 : 1
   B. 1 : 2
   C. 2 : 1
   D. 1 : 4

3. 8 workers can make 4 wardrobes in 12 days. If 15 workers complete the same task in x days, then we have

   A. $\frac{8}{12} = \frac{x}{15}$
   B. $\frac{8}{15} = \frac{x}{12}$
   C. $\frac{8}{15} = \frac{12}{x}$
   D. $\frac{8}{x} = \frac{4}{15}$

4. If the annual savings of Mrs. Sajid for the year 2011 are Rs. 500,000, then the amount of Zakat paid by her will be

   [**Note**: Rate of Zakat is 2.5% of annual saving]

   A. Rs. 12,500
   B. Rs. 15,500
   C. Rs. 25,500
   D. Rs. 50,500

5. If a person earns Rs. 4, 00,000 and 6, 00,000 from two agriculture farms A and B respectively, the amount of Ushr due to be paid by him will be

   [**Note**: Rate of Ushr is 10% of earning]

   A. Rs. 1,00,000
   B. Rs. 60,000
   C. Rs. 40,000
   D. Rs. 20,000

6. If Naureen buys an article costing Rs. 650 for Rs. 585, the discount she has received is

   A. 9%
   B. 10%
   C. 13%
   D. 15%

7. A salesman offered 25% discount on every article. On purchasing an article if Jamal saved Rs. 125, then the price of the article was

   A. Rs. 475
   B. Rs. 500
   C. Rs. 625
   D. Rs. 650

8. What will be the selling price, if 20% discount is offered on the marked price of Rs. 450?

   A. Rs. 360
   B. Rs. 310
   C. Rs. 470
   D. Rs. 480

9. If the principal amount is Rs. 1,000 and the rate of interest is 1% per annum, then after 1 year the markup will be Rs.

   A. 1
   B. 1.1
   C. 10
   D. 10.1

10. A deep freezer costs Rs. 40,000 on lease with down payments of Rs. 4,000. What will be the monthly installment if the remaining amount is to be paid in 20 equal monthly installments?

    A. Rs. 1, 800
    B. Rs. 2, 000
    C. Rs. 2, 200
    D. Rs. 2, 400

11. The instrument used to draw money from a bank account is called a

    I. cheque.
    II. debit card.
    III. deposit slip.

    A. I and III only
    B. I and II only
    C. III only
    D. I only

12. A person insured his car at an assessed market price of Rs. 9, 00,000. The annual premium is 7% of the assessed market price. The annual premium paid is

    A. Rs. 6300
    B. Rs. 7470
    C. Rs. 63000
    D. Rs. 74700

13. Shumaila used a mobile card of Rs. 1000 for recharge and the amount credited to her balance is Rs. 820. The tax deducted was

    A. 13%
    B. 16%
    C. 18%
    D. 20%

14. If Javed earns a yearly basic salary of Rs. 1, 44,000 and receives other allowances of Rs. 2000 per month, then his total annual income is

    A. Rs. 1,20,000
    B. Rs. 1,42,000
    C. Rs. 1,46,000
    D. Rs. 1,68,000

15. If the sales tax charged on a commodity at the rate of 13% is Rs. 130, then the price of the commodity is

    A. Rs. 1000
    B. Rs. 987
    C. Rs. 870
    D. Rs. 831

16. Asad's daily income is Rs.1000 excluding Saturday and Sunday. If on Saturday and Sunday he earns 20% more than the other days, then his weekly income is

    A. Rs. 5200
    B. Rs. 5400
    C. Rs. 7200
    D. Rs. 7400

17. $a^{-\frac{5}{3}}$ can be written in radical form as

    A. $\sqrt[3]{a^5}$
    B. $\sqrt[5]{a^3}$
    C. $\frac{1}{\sqrt[3]{a^5}}$
    D. $\frac{1}{\sqrt[5]{a^3}}$

18. $\log_a y = x$ is possible only when

I. $a > 0$ and $x > 0$
II. $a \neq 1$
III. $y < 0$

A. I only
B. III only
C. I and II only
D. I and III only

19. If $\log_{1+a} a^2 = 1$, then

A. $a^2 = 1$
B. $a^2 = 2 + 2a$
C. $a^2 = 1 + a$
D. $a^2 = 1 - a$

20. The arithmetic mean of – 1 and $\frac{1}{2}$ is

A. $\frac{1}{4}$
B. $-\frac{1}{4}$
C. $-\frac{1}{2}$
D. $\frac{1}{2}$

21. If $a_{n-1} = 2(3)^{n-2}$, then a_n is equal to

A. $2(3)^n$
B. $2(3)^{n-3}$
C. $2(3)^{n-1}$
D. $2(3)^{n+1}$

22. The 10th term of the sequence $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots\ldots$ is

A. $\left(\frac{1}{2}\right)^9$

B. $\left(\frac{1}{2}\right)^{10}$

C. $\left(\frac{3}{2}\right)^9$

D. $\left(\frac{3}{2}\right)^{10}$

23. If $K = \{2,3\}$, $L = \{2,3\}$ and $K \cap M = L$, then which of following sets is true for set M ?

I. $\{2,3\}$
II. $\{1,3,5\}$
III. $\{1,2,3,6\}$

A. I only
B. II only
C. I and III only
D. II and III only

24. If $E \cup F = \{a,b,c,d,e\}$ and $E \cap F = \{b,d,e\}$, then $E \Delta F$ is equal to

A. $\{e\}$
B. $\{a,c\}$
C. $\{b,d\}$
D. $\{a,b,c,d,e\}$

25. For any set *E*, which of the following conditions is FALSE?

A. $E \cup E = E$
B. $E \cap E = E$
C. $E \cup \phi = E$
D. $E \cap \phi = E$

26. Which of the following statements is TRUE?

A. Every relation is a function
B. Every function is a one-one correspondence
C. A one-one correspondence is a bijective function
D. An onto and a one-one function are surjective functions

27. The point which satisfies the equation $\frac{x}{2} + y = -1$ is

A. $(2,-2)$
B. $(1,-2)$
C. $(-2,2)$
D. $(-2,1)$

28. If $37°C = 98.6°F$, then $18.5°C$ is equal to

A. $49.6°F$
B. $4.96°F$
C. $4.93°F$
D. $49.3°F$

29. For the given cumulative frequency table, the value of k will be

Class Interval	Frequency	Cumulative Frequency
5 – 10	3	
10 – 15	?	k
15 – 20	5	
20 – 25	7	
25 – 30	3	20

A. 5
B. 6
C. 15
D. 18

30. If the minimum value and the range of a data set are 10 and 40 respectively, then the maximum value will be

A. 10
B. 30
C. 40
D. 50

END OF PAPER

Please use this page for rough work